

KUMLINGE KK, Finland

WMO: 027900

Lat: 60.258N

Lon: 20.747E

Elev: 23

StdP: 101.05

Time Zone: 2.00 (EUE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.7	-11.0	-17.3	0.8	-13.7	-14.4	1.1	-10.3	14.4	1.2	12.7	0.2	3.9	80	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	24.2	19.2	22.6	18.1	21.2	17.3	20.4	23.0	19.1	21.8	18.0	20.4	4.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.2	22.0	18.1	13.0	20.8	17.0	12.1	19.5	58.7	23.0	54.5	21.7	51.1	20.4	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.1	9.9	DB	-14.8	25.7	4.5	1.9	-18.0	27.0	-20.6	28.1	-23.1	29.2	-26.4	30.6	(4)
(5)			WB	-15.4	20.4	4.2	1.7	-18.5	21.6	-21.0	22.6	-23.3	23.5	-26.4	24.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-1.6	-2.4	-0.5	3.9	9.4	13.5	17.7	17.2	13.3	7.9	4.2	1.5	(6)
(7)		DBStd	7.64	3.96	4.78	3.78	2.77	3.28	2.76	2.71	2.34	2.37	2.94	3.14	3.58	(7)
(8)		HDD10.0	1790	358	348	326	183	51	4	0	0	4	77	176	263	(8)
(9)		HDD18.3	4155	617	582	584	432	277	146	47	48	152	324	426	522	(9)
(10)		CDD10.0	721	0	0	0	1	33	109	238	225	103	11	0	0	(10)
(11)		CDD18.3	42	0	0	0	0	0	1	26	14	0	0	0	0	(11)
(12)		CDH23.3	89	0	0	0	0	1	3	68	18	0	0	0	0	(12)
(13)		CDH26.7	4	0	0	0	0	0	0	4	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.1	5.7	5.1	5.1	4.8	4.5	4.6	4.4	4.7	5.0	5.5	5.8	6.3	(14)
(15)	Precipitation	PrecAvg	835	87	64	53	42	43	61	74	79	70	83	86	94	(15)
(16)		PrecMax	1040	147	136	98	71	91	112	138	144	142	170	197	165	(16)
(17)		PrecMin	618	13	2	10	6	10	25	4	27	20	12	13	18	(17)
(18)		PrecStd	101	37	33	21	18	21	22	33	30	37	38	37	37	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	5.5	8.5	15.1	21.2	23.1	27.1	25.4	19.7	14.7	10.2	8.3	(19)
(20)			MCWB	5.0	3.5	5.3	9.2	14.1	16.0	19.9	20.3	16.1	12.8	9.4	6.0	(20)
(21)		2%	DB	5.0	4.1	6.5	11.8	18.7	21.0	25.2	23.3	18.2	13.3	9.2	7.0	(21)
(22)			MCWB	4.2	2.8	4.0	7.4	13.1	15.2	19.7	18.8	15.3	12.0	8.5	5.3	(22)
(23)		5%	DB	4.2	3.3	4.9	9.6	16.4	19.4	23.5	21.9	17.2	12.4	8.6	6.2	(23)
(24)			MCWB	3.5	2.4	3.1	6.0	12.1	14.5	18.9	17.8	14.8	11.2	7.9	5.0	(24)
(25)		10%	DB	3.3	2.5	3.8	8.0	14.4	17.8	22.0	20.6	16.3	11.6	7.9	5.4	(25)
(26)			MCWB	2.5	1.8	2.4	5.1	10.7	13.8	18.3	17.1	14.3	10.4	7.1	4.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	3.9	5.7	9.7	15.6	17.7	22.0	21.9	16.9	13.6	9.8	7.2	(27)
(28)			MCDWB	5.7	4.6	8.0	14.0	19.1	21.1	25.2	24.4	18.5	14.3	10.0	7.8	(28)
(29)		2%	WB	4.4	3.1	4.4	7.9	13.9	16.3	20.9	19.9	15.9	12.3	8.8	6.1	(29)
(30)			MCDWB	4.8	3.7	5.9	11.3	17.7	19.4	23.4	22.7	17.6	13.0	9.1	6.6	(30)
(31)		5%	WB	3.6	2.5	3.4	6.5	12.4	15.4	19.9	18.7	15.3	11.5	8.0	5.3	(31)
(32)			MCDWB	4.1	3.1	4.5	9.0	15.9	18.3	22.4	20.8	16.7	12.2	8.5	5.8	(32)
(33)		10%	WB	2.6	1.8	2.6	5.3	11.1	14.3	18.8	17.8	14.7	10.7	7.2	4.6	(33)
(34)			MCDWB	3.2	2.4	3.6	7.4	14.2	17.0	21.6	19.8	16.0	11.5	7.7	5.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.2	3.6	4.5	5.3	5.9	5.3	4.7	4.3	3.6	3.0	2.6	2.9	(35)
(36)			MCDBR	3.1	3.3	5.3	8.1	9.3	8.0	6.9	5.9	4.4	3.3	2.6	3.1	(36)
(37)			MCWBR	3.0	3.0	3.7	4.9	5.0	3.8	3.2	2.8	2.6	2.7	2.7	3.1	(37)
(38)		5% WB	MCDWR	3.1	3.1	5.0	7.7	8.8	7.0	6.0	5.0	4.1	3.2	2.5	2.9	(38)
(39)			MCWBR	3.1	2.8	3.7	5.0	5.2	3.8	3.2	3.0	2.8	2.8	2.7	3.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.310	0.329	0.368	0.366	0.357	0.386	0.376	0.347	0.332	0.300	0.386	(40)	
(41)		taud	2.177	2.251	2.304	2.329	2.424	2.492	2.437	2.452	2.516	2.413	2.219	2.649	(41)	
(42)		Ebn at Noon	486	680	796	829	861	875	836	816	783	658	444	341	(42)	
(43)		Edh at Noon	41	68	89	104	102	97	101	92	73	59	40	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.23	0.83	2.39	4.04	5.54	6.11	5.68	4.42	2.70	1.15	0.31	0.14	(44)	
(45)		RadStd	0.03	0.15	0.27	0.38	0.47	0.41	0.51	0.38	0.30	0.17	0.06	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (45 neighbors)		+0.36	N/A	N/A	N/A	N/A	N/A	-110	N/A	N/A	N/A	(47)

Nomenclature: See separate page